Quiz 4

September 18, 2026 Name: _____

Show your work. **Circle your answer.** **Total: 10 points**

1. Quadratic equation and vertex [6 points]

Let

$$f(x) = x^2 - 4x - 5.$$

(a) Find the discriminant of $f(x) = 0$. State the number and type of solutions. **[2 points]**

(b) Solve $f(x) = 0$ using the quadratic formula. **[2 points]**

(c) Find the vertex, axis of symmetry, and range of f. **[2 points]**

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2. Zeros and end behavior **[4 points]**

Let

$$P(x) = -(x+1)(x-2)^2.$$

(a) Find the degree and leading coefficient. **[1 point]**

(b) Find each real zero and its multiplicity. State whether the graph crosses or touches the x-axis there. **[2 points]**

(c) Write the limits of $P(x)$ as $x \to -\infty$ and $x \to \infty$. **[1 point]**